

Student Behavior Monitoring in Experimental Exams: Anti-Exam Model and STEAM-Exam Dataset

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Abstract

This paper introduces the Anti-Exam model and the STEAM-Exam dataset to enhance behavior recognition during AI-assisted scoring of junior high school practical examinations, promoting exam integrity, fairness, and student's personal development. The Anti-Exam model combines MODNet portrait matting with the YOLOv9 object detection algorithm, leveraging images from a real-time AI-assisted scoring device equipped with three cameras to simulate a realistic examination environment. The STEAM-Exam dataset includes both MODNet-processed images and those with natural backgrounds, demonstrating varying processing speeds based on background complexity. Evaluation results reveal that the YOLOv9 model achieves the highest precision (98.7%), recall (98.7%), mAP50 (99.4%), and mAP50:95 (98.6%) among YOLOv7 and YOLOv8 models. Processing speeds differ between images with backgrounds (37.4ms per image for inference) and plain background images (35.1ms per image for inference). This research underscores the importance of AI in ensuring exam fairness and transparency, addressing challenges such as occlusion and cheating detection. The STEAM-Exam dataset is a critical resource for advancing AI-assisted educational assessments, aligning with China's policy to include practical examination results in entrance scores. Our findings suggest significant potential for AI to revolutionize educational assessments, ensuring fairness and integrity while supporting student development.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision tasks; Activity recognition and understanding.

Keywords

Anti-Exam, YOLOv9, MODNet, STEAM-Exam dataset, Examination room monitoring, Experimental exams

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1 INTRODUCTION

With the continuous promotion of educational evaluation reform, the Ministry of Education in China has established a policy that, by 2023, the practical examination results of physics, chemistry, and biology in junior high schools will be included in the entrance examination scores. This policy has been rapidly implemented in various regions, including Shanghai, Shenzhen, Guangdong, Zhejiang, Inner Mongolia, and Heilongjiang [2]. It emphasizes the importance of scientific practice skills among middle school students and has become a key indicator for evaluating academic performance. AI-based invigilation schemes have gained rapid popularity with the increasing demand to detect and prevent student cheating [1]. These artificial intelligence (AI) systems play a significant role in ensuring the fairness of experimental examinations, impacting the future educational trajectory of over 50 million middle school students nationwide. AI has revolutionized numerous fields, including education. Traditional methods of evaluating student performance, particularly in practical examinations, are often time-consuming and fail to capture the full spectrum of a student's behavior and capabilities. Additionally, these methods are inadequate for monitoring and preventing academic misconduct during examinations. The need for advanced, real-time monitoring and evaluation systems has led to the development of AI-assisted examination tools. Despite the growing interest and application of AI in educational assessments, there is a significant lack of publicly available datasets tailored to the practical examination environments of junior high schools. This gap presents a considerable challenge for researchers and practitioners aiming to enhance the integrity and fairness of exams through AI.

In the standardized test room environment, there are usually surveillance cameras before and after the test room, and the monitoring range of these cameras can cover every examinee's test position. The test comes in various forms, from traditional paper-and-pencil quizzes to computer-based tests. During the test, the candidate maintains the correct sitting posture and ensures, that no objects on the table may block the view. Due to the simplicity of the test room environment, AI detection technology has been widely used to monitor test behavior in standardized exam rooms, effectively identifying cheating and abnormal behaviors of candidates, such as turning back, sidelong glancing, sharing answers, communicating with others, consulting books or mobile phones [3], and walking around, as illustrated in Figure 1.

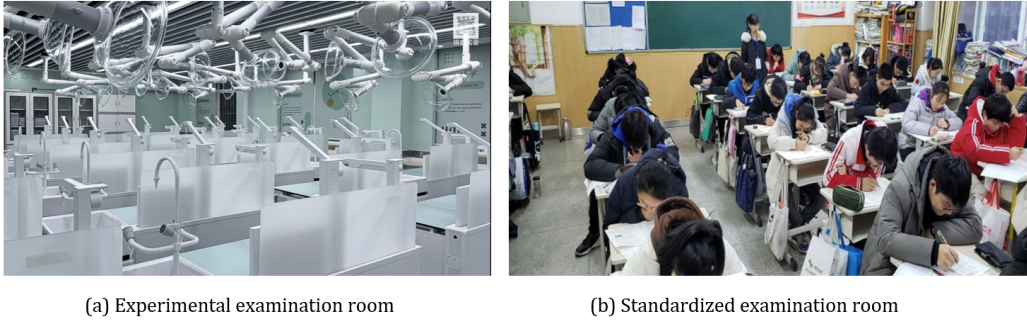


Figure 1: Comparison of environment between (a) experimental and (b) standardized test rooms

In physics, chemistry, biology, and other experimental examinations, invigilation is still mainly dependent on manual monitoring. This is because the environment of the laboratory examination room is highly complex and variable: the roof is equipped with electricity and ventilation facilities, the exam bed is arranged with a variety of instruments and equipment, and to prevent liquid spillage, the exam room is sometimes set up baffles. In this environment, candidates may need to stand, sit, or move between test seats, making it difficult for traditional surveillance cameras to monitor all aspects of the exam, thus being unable to effectively identify abnormal behavior. In addition, current AI scoring and experimental operation monitoring systems require expensive GPU servers for schools to run. Especially when there is a problem in the experimental network, the single-machine system cannot complete the real-time detection of AI. Therefore, there is an urgent need for a new solution to assist manual proctors, solve the problems of monitoring cameras being occluded, and achieve high accuracy and speed test behavior detection in a low computing power environment, to accurately identify the abnormal experimental operation behavior of candidates, and create a fair and transparent environment for the experimental examination.

In the research field of cheating behavior analysis in standardized examinations, existing algorithms focus on technical frameworks such as CNN, RNN, VGG, Transformer, and so on. Action recognition technique based on the combination of LSTM, VGG-19, Faster R-CNN, Mask R-CNN, and Vision Transformer architectures is proposed in [6]. This method was designed to detect students' behaviors such as copying or exchanging papers among students during examinations. Experimental results show that the Vision Transformer can identify fraud with over 98% accuracy, aiding teachers in managing examinations and promoting fairness and transparency in education [6].

A study provides an in-depth discussion of practical experiences in constructing chemistry and physics experimental manipulation exams in middle schools, as well as current challenges and future directions, particularly highlighting the importance of AI in the education field [7]. Another analysis examines the unfairness in middle school science experiment operation tests from the perspective of the examinees, proposing measures to ensure fairness such as adopting a unified scoring standard and monitoring system [8]. Reflecting on the practice of implementing middle school academic-level experimental operation skills tests, researchers have deeply

analyzed existing problems and suggested artificial intelligence use to assist in the examination, offering a new perspective for improving fairness and efficiency [1]. In the field of cheating detection in experimental operation examinations in middle schools, the scarcity of literature resources indicates that research is still in its preliminary stages and requires more extensive academic communication and further in-depth study. Moreover, while there have been attempts to introduce standardized test room modalities into the experimental manipulation test, the complexity of the test environment—such as background and light changes, and the dynamic interaction between candidates and proctors—poses challenges for accurately identifying abnormal behaviors in the test [4].

Our primary contribution to this research is the creation of the STEAM-Exam dataset, a comprehensive collection of data designed to address the challenges of behavior recognition during junior high school practical examinations. This dataset was compiled using real-time data collected via the AI-assisted device. The device, widely used in schools for AI-based scoring, is equipped with three cameras (front, top, and side) that continuously monitor students during their practical exams. By leveraging object detection and action recognition technologies, the device automatically assigns scores to students. However, recognizing and preventing cheating and other undesirable behaviors remain critical challenges.

To tackle these challenges, we integrated MODnet [13] for video background matting and YOLOv9 [9] for object detection. The performance of these models was meticulously evaluated using metrics such as mean average precision (mAP@50 and mAP@50-95), precision, and recall. The STEAM-Exam dataset enhances the integrity of the examination process by enabling robust behavior recognition, thus creating a more secure and fair examination environment. This dataset is a significant advancement in developing intelligent educational assessment tools, offering a comprehensive solution for real-time monitoring and evaluation of student performance during practical exams.

This paper presents the Anti-Exam model, which integrates MODnet and YOLOv9, demonstrating superior performance compared to previous models such as YOLOv7 and YOLOv8. Our findings indicate that the YOLOv9-E model achieved the highest precision, recall, and mAP scores, highlighting its effectiveness in recognizing student behavior during exams. Furthermore, we provide a detailed comparison of processing speeds for images with

backgrounds versus plain background images, underscoring the efficiency of our approach.

Overall, our research significantly contributes to the field of AI in education by providing a robust framework for ensuring exam integrity and fairness. The STEAM-Exam dataset and Anti-Exam model offer valuable tools for real-time, AI-assisted examination scoring, paving the way for future advancements in educational assessments.

2 RELATED WORK

Behavior recognition of students during junior high school practical examinations presents a significant challenge due to the lack of publicly available datasets tailored to practical examination environments. Our primary contribution is the creation of a dataset named the STEAM-Exam dataset. This dataset was compiled using real-time data collected via an AI-assisted device, which is widely utilized in numerous schools for AI-based scoring during practical examinations. This device is equipped with three cameras (front, top, and side) that continuously monitor students as they undertake their practical exams. Leveraging object detection and action recognition technologies, the device automatically assigns scores to students. The subsequent critical challenge involves behavior recognition to prevent cheating and other undesirable student behaviors during exams [5]. To address this, we have integrated MODNet for video background matting and YOLOv9 for object detection. We have meticulously evaluated the model's performance using metrics such as mean average precision (mAP@50 and mAP@50-95), precision, and recall. This dataset enhances the integrity of the examination process by enabling robust behavior recognition. We aim to create a more secure and fair examination environment by employing advanced AI techniques. The STEAM-Exam dataset thus represents a significant step forward in developing intelligent educational assessment tools, offering a comprehensive solution for real-time monitoring and evaluation of student performance during practical exams.

2.1 Behavior Detection

Student classroom behavior recognition has many challenges such as occlusion, complex environment, and density of students to deal with challenge. The author proposed a student behavior detection in the classroom system based on Improved YOLOv8. The Res2Net module is integrated with the C2f module in YOLOv8 and creates a novel C2f_Res2block module. This model handles multi-scale scenarios. The efficient multi-head self-attention (MHSA) and the multi-scale attention module (EMA) mechanism module were introduced to strengthen performance [12].

2.2 Improving Portrait Video Matting via Background Restoration

The proposed approach in this work addresses the challenge of portrait video matting by introducing a background restoration module (BRM) that dynamically recovers the background image from the input video, eliminating the need for additional user-provided priors like trimaps or background images [14]. The BRM is lightweight and integrates easily into existing models. Combining BRM with the MODNet image matting model creates the new MODNet-V,

which has only one-third of the parameters of MODNet but delivers comparable or superior performance. MODNet-V can be trained end-to-end on a single NVIDIA 3090 GPU. Additionally, a new patch refinement module (PRM) is introduced to enable MODNet-V to handle high-resolution videos while remaining efficient. The results show that MODNet-V significantly improves the quality and stability of matting compared to previous state-of-the-art methods like MODNet [14].

2.3 Real-Time Trimap-Free Portrait Matting via Objective Decomposition

Existing portrait matting methods often require costly auxiliary inputs or are computationally expensive, making them unsuitable for real-time applications. The proposed lightweight Matting Objective Decomposition Network (MODNet) addresses these issues by using a single input image and optimizing sub-objectives simultaneously through explicit constraints [13]. MODNet includes two novel techniques: The Efficient Atrous Spatial Pyramid Pooling (e-ASPP) module for multi-scale feature fusion and a self-supervised sub-objective consistency (SOC) strategy to adapt to real-world data and address domain shift problems. MODNet can be trained end-to-end and runs at 67 frames per second on a 1080Ti GPU, outperforming prior trimap-free methods on various benchmarks and real-world data. However, it struggles with videos having strong motion blurs due to the lack of temporal information [13].

2.4 Spatio-Temporal Network and Multi-Model Fusion

Deep learning models require manual labeling and annotation for object detection, pose estimation of even facial expression analysis to recognize the student behavior in the classroom or during an exam [11] proposed a method to extent image dataset to detect student behavior and analysis. Spatio-Temporal Network and Multi-Model Fusion technique is used. SCB-ST-Dataset4 [11] focuses on three behaviors; reading, writing, and hand raising. Deep Sort, YOLOv7, SynergyNet, and Facial Expression models' fusion to detect and analyze the student behavior in the classroom.

3 RESEARCH METHODOLOGY

3.1 AI-Assisted Exam

In recent decades, AI-assisted examinations have gained popularity worldwide, particularly in China. Traditional exam evaluation is time-consuming and often fails to capture the full spectrum of a student's performance during the exam. Additionally, it does not efficiently monitor student behavior or distractions, such as focus, engagement, and fairness. AI-assisted practical exams play a significant role in addressing these challenges. However, there is a lack of publicly available datasets to observe student behavior specifically in the context of junior high school academic-level practical examinations, presenting a significant challenge for AI researchers in the educational field. To address this issue, we have developed a student behavior detection dataset named the "STEAM-Exam dataset" for real-time AI-assisted examinations.

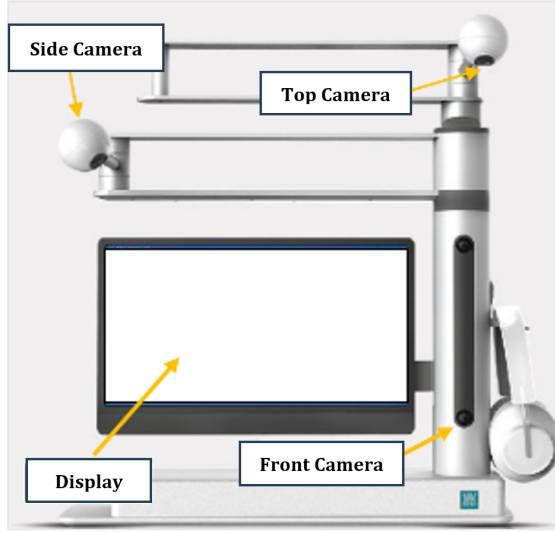


Figure 2: AI-Assisted device for video data collection

3.2 STEAM-Exam dataset

STEAM-Exam dataset is designed to recognize student behavior during junior high school practical examinations to prevent misconduct. The images in the STEAM-Exam dataset were captured using a real-time AI-assisted scoring device as shown in Figure 1. This device is employed for AI scoring in junior high school STEAM examinations using three AI-assisted cameras (front, top, and side), as illustrated in Figure 2. It consisted of an AI development module, designed for real-world AI, Internet of Things (IoT), and edge computing applications. This module features an octa-core 64-bit processor and an ARM Mali-G610 MP4 GPU, and it supports up to 32GB RAM, 8K video encoding and decoding, and multiple video input/output interfaces.

The videos were collected using the front, top, and side cameras of the RK3588 AI-assisted device in a real examination environment. Images were then extracted from the videos at the 2 frames per second rate. The STEAM-Exam dataset consists of 4,150 images from the front, top, and side camera views. All images are annotated and labeled using the "labellmg" labeling tool. The dataset also includes plain background images as shown in Figure 3, that were obtained using the MODnet model to increase model accuracy during training and testing by focusing on the target object.

To ensure accurate detection of student behavior and to prevent unfair practices during examinations, we have created specific labels such as "Looking at the phone," "Reading a book or notes," "Writing," and "Playing with a watch." Additionally, to address other potential distractions or behaviors such as sleeping during the exam, we have included labels like "Playing with fire," "Playing with a pen - Taking a pen," and "Sleeping during the exam" as shown in Figure 3.

4 PROPOSED MODEL

In our proposed model we have integrated MODnet for background matting which is real-time trimap-free portrait matting via objective

decomposition, which uses the Efficient Atrous Spatial Pyramid Pooling (e-ASPP) module for multi-scale feature fusion and a self-supervised sub-objective consistency (SOC) strategy technique, and YOLOv9 which is highly accurate with high-speed object detection algorithm as compared to other object detection models such as YOLOv7 [10] and YOLOv8 [15]. Deep learning models often lose information during data processing. YOLOv9 uses Programmable Gradient Information (PGI) to address information loss and improve gradient calculation. A new network architecture, Generalized Efficient Layer Aggregation Network (GELAN), is designed based on PGI and it achieves better performance than existing methods using lightweight models. YOLOv9 achieves better performance with fewer parameters and computations compared to YOLOv8 [15] and YOLOv7 [10].

In this paper, we utilize YOLOv9 for real-time object detection and tracking of students within examination videos, combined with background matting using the MODnet technique. This approach isolates students from background distractions, allowing the model to focus on and recognize student behaviors during the examination. As illustrated in Figure 4, the process begins with data pre-processing. In the initial stage, video data is collected and subsequently extracted into frames. These frames are then processed through the image matting stage, which removes background elements from the images. The resulting background-matted images are used for training and testing the YOLOv9-based model.

Where,

I , is input image

Sp , MODNet predicts portrait semantics

dp , boundary details

αp , final alpha matte

S , D , and F , three interdependent branches

By isolating students from their surroundings, the model can accurately target and monitor student behavior without interference from extraneous background details. This enhanced focus improves the reliability of behavior recognition during examinations, ensuring that the model can effectively identify actions such as using a phone, reading notes, or any other behavior that might indicate misconduct or distraction. Furthermore, the integration of YOLOv9 with MODnet background matting optimizes the model's performance by concentrating computational resources on relevant features, thereby enhancing both accuracy and efficiency. This methodology not only ensures precise behavior detection but also facilitates real-time analysis, which is crucial for maintaining the integrity of the examination process.

4.1 Image Matting Elimination

The proposed model removes the background of the video frames before performing the behavior recognition of the students. To remove the background from the video, we have used the trimap-free MODNet model for portrait matting in real time. In Figure 4(b), the "image matting eliminates" is a "trimap-free MODNet model". It eliminates the background of each frame of the video and make plain backgrounded video before performing the YOLOv9 object detection algorithm. The input image is processed through three branches, high resolution branch (D), low resolution branch (S), and a semantic-detail fusion branch (F). The high-resolution branch

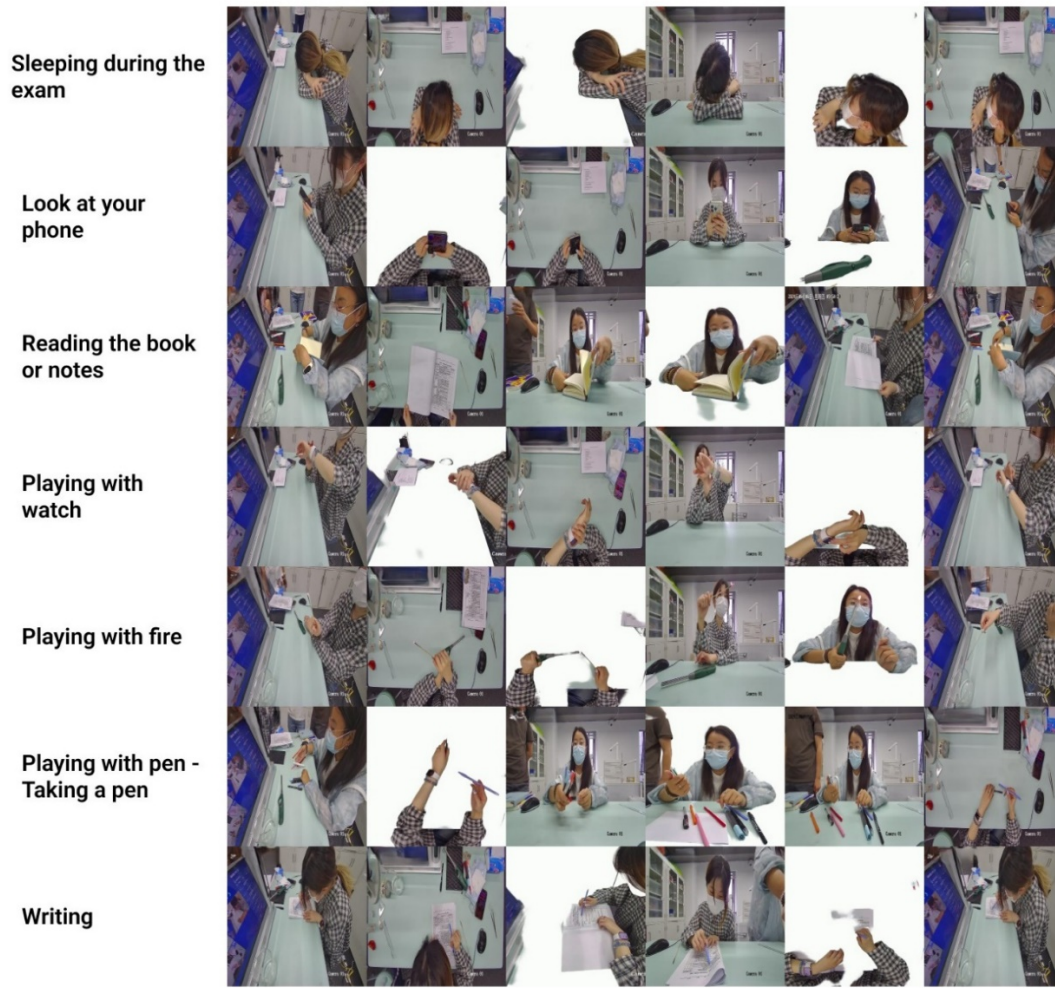


Figure 3: STEAM-Exam dataset images of students' behavior

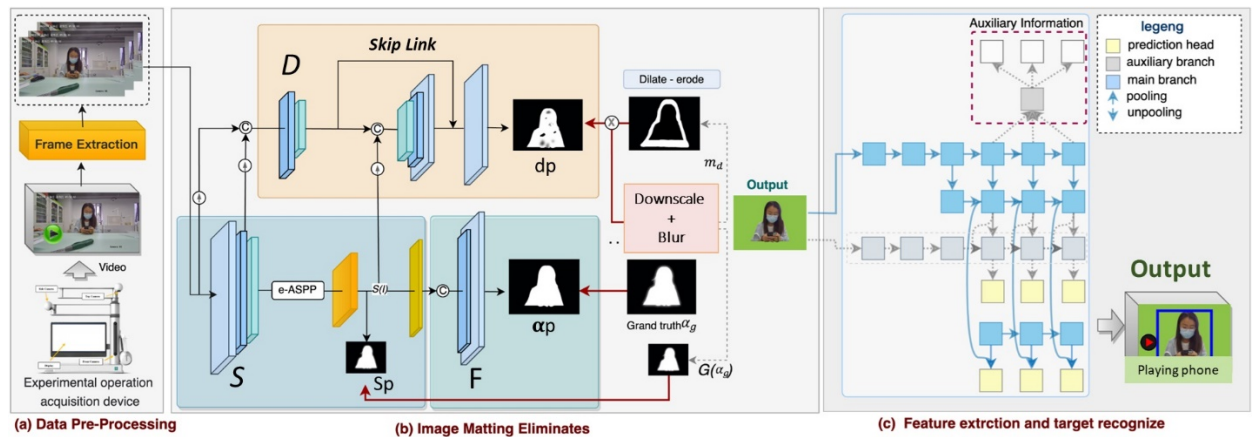


Figure 4: Proposed Model Architecture (Anti-Exam Model)

is used to predict detailed features of the image. The “skip link” allows the model to retain high-resolution details. The “Details dp” is the output of the branch, which defines the details of the image necessary for accurate matting. The low-resolution branch, it focuses on semantic estimation, processing the image at a lower resolution to capture broader context and semantics. It includes an enhanced Atrous Spatial Pyramid Pooling (e-ASPP) module, which helps in capturing multi-scale context by applying atrous convolutions with different rates. The output of this branch is the semantics (Sp), which represent the broader semantic features of the image. The semantic-detail fusion branch fuses the details from the high-resolution branch and the semantics from the low-resolution branch. The fusion helps in integrating fine details with the broader semantic context to produce a comprehensive understanding of the image. The output here is the alpha matte (αp), which is the refined prediction of the foreground object. Post-processing steps refine the alpha matte by applying morphological operations (dilation and erosion) to smooth and clean the predicted foreground. The ground truth alpha matte is used for training, representing the ideal separation of foreground and background. The final output is the refined image with the foreground object accurately separated from the background.

4.2 Impact of Image Background on Inference Time

Inference time for YOLOv9 object detection can be influenced by various factors, including the complexity of the image background. The background in the image introduces visual elements that the model needs to process, this can increase the computational load as the model needs to distinguish between the object of interest and objects in the background. The model will analyze the more features of the image in the background that are not required. Due to the complex background in the training images, during model evaluation and testing the model can generate false positive results, which leads to low accuracy and longer inference time as the model needs to handle and filter out these incorrect detections. After analyzing and evaluating the results of our model, we found that inference time with a plain background is short as compared to a complex background and trained model evaluation matrix results are highly accurate as shown in the results section. The model achieved 99.4 mAP@50 (%).

4.3 Feature Extraction and Target Recognition

After the image is matted using the MODnet technique, the output image is used in training and testing the Anti-Exam model based on YOLOv9. Figure 4(c) “Feature Extraction and Target Recognition,” outlines the process by which features are extracted from processed images to recognize specific targets or behaviors during examinations. The clear, processed image from the previous stage is fed into this module, where an auxiliary branch assists the main branch in enhancing feature extraction performance. The main branch, comprising interconnected layers likely from a convolutional neural network (CNN), uses pooling and unpooling operations to capture features at multiple scales. The interconnected structure ensures the capture of both high-level and low-level features. The prediction head then generates the final output predictions by interpreting

these feature maps, labeling detected behaviors or objects. The final output, such as a student “playing phone” during an exam, is visually highlighted, aiding in monitoring and analyzing student actions. This part demonstrates how the system combines auxiliary information with detailed feature extraction to accurately recognize and label behaviors or objects, ensuring effective examination monitoring and evaluation.

Overall, our approach leverages advanced object detection and background matting techniques to create a robust system for monitoring and evaluating student behavior during exams, providing a significant improvement over traditional methods. This system’s ability to isolate and analyze individual student actions in real time represents a valuable tool for educational institutions aiming to uphold exam integrity and fairness.

5 EXPERIMENTAL SETUP

5.1 Experiment environment

The experiment was conducted in an Anaconda virtual environment utilizing Python version 3.9. For deep learning tasks, we employed torch version 2.3.1 on a Linux operating system with CUDA 11.4 support (NVIDIA GeForce RTX 2060, 5924MiB). Essential libraries such as Ultralytics and OpenCV were integral to our implementation.

5.2 Dataset and Model Training

Our experiments utilized the STEAM-Exam dataset, which we partitioned into training and validation sets at a ratio of 4:1, comprising images and corresponding .txt labels. The model was trained over 300 epochs with a batch size of 4 and image dimensions set to 640x640 pixels. We leveraged pre-trained models to enhance training efficacy.

5.3 Evaluation Matrix

To evaluate the performance of the model we have used the confusion matrix and it makes easier to understand how well the model is performing. It is used to calculate several important metrics to evaluate the performance of the model, including Accuracy, Precision (Eq. 1), Recall (Eq. 2), and F1 score (Eq. 3). To evaluate the model’s Precision and Recall the Average Precision (AP, Eq. 4) and Mean Average Precision (mAP, Eq. 5) have been calculated.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$AP_i = \int_0^1 P(r) dr \quad (4)$$

$$\text{mAP} = \frac{1}{n} \sum_i (AP_i) \quad (5)$$

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (6)$$

To calculate the precision, we have (Eq. 1) in which we have divided the True Positive (TP) values and the sum of True Positive

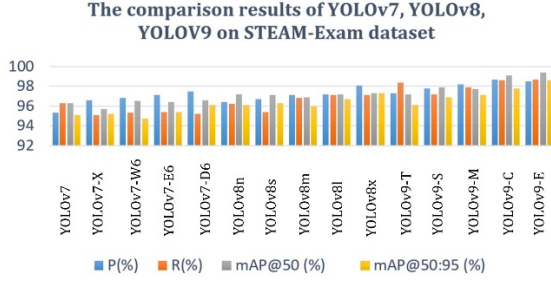


Figure 5: The comparison results of YOLOv7, YOLOv8, and YOLOv9 on STEAM-Exam dataset

(TP) and False Positive (FP). It will measure the correctness of positive predictions. To calculate Recall in (Eq. 2) we have divided TP values with the sum of TP and FN values, which measures the model’s ability to find all positive instances. F1 score is calculated in (Eq. 3) which is used to measure the harmonic mean of precision and recall. It divides the multiplication of Precision and Recall with the sum of Precision and Recall. AP (Eq. 4) is used to measure the Average Precision under the Precision-Recall curve and mAP (Eq. 5) is the mean of Average Precision values measured for each class. Intersection over Union (IoU) is metric is used to evaluate the object detection accuracy. It measures the overlap between the predicted bounding box and ground truth bounding box (Eq. 6). We have measured the mAP of the model and each class at the 50% IoU threshold and the mAP at 50:90% IoU threshold.

6 EXPERIMENTAL RESULTS ANALYSIS AND COMPARISON

We trained our proposed Anti-Exam model which is based on YOLOv9 and MODnet integration for video background matting and behavior recognition, STEAM-Exam dataset. The results are evaluated using a confusion matrix by calculating Precision Metrics, Recall Metrics, F score, mAP50, and Map50-95. Our dataset is trained and evaluated on YOLOv7, YOLOv8, and YOLOv9 and the detailed experiments results are shown in Table 1.

From Table 1 and Figure 5, the results of the experiments show-case the performance metrics of YOLOv7, YOLOv8, YOLOv9 various models, we focused on their precision (P), recall (R), and mean average precision (mAP) scores at different IoU thresholds (50% and 50:95%). In YOLOv7, the YOLOv7-D6 model achieves the highest performance (96.1). The YOLOv8 series further improves the performance on these metric and YOLOv8x is achieving the highest precision and mAP (97.3). The YOLOv9 has the most advanced achievement due to its high speed and accuracy. The YOLOv9-C and YOLOv9-E are achieving the highest precision (98.7 %), recall (98.7), mAP50 (99.4), and mAP50:95 (98.6 %). These results value illustrate the trend of performance increase in newer models, where YOLOv9 is achieving the highest performance results on our STEAM-Exam data.

The performance results of the YOLOv9-E model achieved the highest performance among other models so we have considered

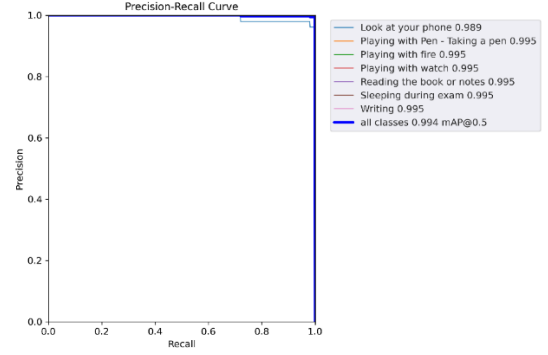


Figure 6: Precision-Recall (PR) curve of STEAM-Exam dataset’s YOLOv9-E for each class

this model to compare the STEAM-Exam dataset’s class performance based on the YOLOv9-E model. From the Table 2, we can observe the comparison results of each class of STEAM-Exam data which is trained on the YOLOv9-E pre-trained model. The results are focusing on P (%), R (%), mAP@50 (%), and mAP@50:95 (%). Overall, the model performs exceptionally well across all classes, with an aggregate precision of 97.8%, recall of 99.7%, mAP@50 of 99.4%, and mAP@50:95 of 94.1%. If we observe individual class such as “Look at your book” and “playing with pen – taking pen” maintain high precision and recall but slightly lower mAP@50:95 (89.1 %). The “Writing” class achieved highest mAP@50:95 of 96.9. Overall, the results highlight the model’s robustness and consistent performance across all classes of our STEAM-Exam dataset.

Figure 6 shows that the model exhibits exceptional performance across all classes, demonstrated by high precision and recall metrics. The model achieves an impressive overall mAP@0.5 score of 0.994, underscoring its robustness and accuracy in behavior detection tasks. The proximity of the curve to the top right corner signifies that the model maintains high precision even as recall increases, indicating effective handling of the trade-off between these two metrics. Furthermore, the close clustering of the individual class curves around the overall performance curve indicates consistent performance across different classes, with minimal variance. This consistency suggests that the model is equally adept at accurately classifying various classes, highlighting its generalizability and reliability. We trained a model on the STEAM-Exam dataset using the YOLOv9 algorithm, achieving an impressive 99.4% mean Average Precision at 50% Intersection over Union (mAP@50%). This model is integrated with the MODnet model for background matting in images.

For student behavior recognition, the YOLOv9 object detection algorithm is employed. The STEAM-Exam dataset includes images both with and without backgrounds. Figures 7 and 8 illustrate the student behavior recognition testing results of the proposed model. Figure 7 shows results for images with backgrounds, while Figure 8 shows results for images with plain backgrounds.

The model demonstrates high accuracy in recognizing student behavior across all seven classes in the STEAM-Exam dataset: “Looking at your book,” “Playing with a pen or taking a pen,” “Playing

Table 1: The comparison results of YOLOv7, YOLOv8, YOLOv9 on STEAM-Exam dataset

Models	P (%)	R(%)	mAP@50 (%)	mAP@50:95 (%)
YOLOv7	95.3	96.3	96.3	95.1
YOLOv7-X	96.6	95.1	95.7	95.2
YOLOv7-W6	96.8	95.3	96.5	94.7
YOLOv7-E6	97.1	95.4	96.4	95.4
YOLOv7-D6	97.5	95.2	96.6	96.1
YOLOv8n	96.4	96.2	97.2	96.1
YOLOv8s	96.7	95.4	97.1	96.3
YOLOv8m	97.1	96.8	96.9	96
YOLOv8l	97.2	97.1	97.2	96.7
YOLOv8x	98.1	97.1	97.3	97.3
YOLOv9-T	97.3	98.4	97.2	96.1
YOLOv9-S	97.8	97.2	97.9	96.9
YOLOv9-M	98.2	97.9	97.7	97.1
YOLOv9-C	98.7	98.6	99.1	97.8
YOLOv9-E	98.5	98.7	99.4	98.6

Table 2: The comparison results of the STEAM-Exam dataset’s YOLOv9-E for each class

Class	P (%)	R(%)	mAP@50 (%)	mAP@50:95 (%)
All Classes	97.8	99.7	99.4	94.1
Look at your phone	96.4	98.3	98.9	89.5
Playing with pen – taking pen	99.8	99.8	99.5	90.4
Playing with fire	99.5	99.5	99.5	93.9
Playing with watch	97.2	99.3	99.5	98.4
Reading the book or notes	99.3	99.4	99.5	98.9
Sleeping during exam	96.8	99.7	99.5	89.1
Writing	95.1	99.2	99.5	98.3

with fire,” “Playing with a watch,” “Reading a book or notes,” “Sleeping during the exam,” and “Writing.” This high level of accuracy is observed in both types of images, those with backgrounds and those with plain backgrounds. The model’s processing speed varies slightly depending on the background complexity of the images. Images with backgrounds, the model’s processing speed is 0.3ms for pre-processing, 37.4ms for inference, and 1.5ms for Non-Maximum Suppression (NMS) per image. In contrast, for plain background images, the processing speed is 0.2ms for pre-processing, 35.1ms for inference, and 0.7ms for NMS per image. These results indicate that the inference time is shorter for images with plain backgrounds compared to those with complex backgrounds.

Overall, the evaluation metrics of the trained model demonstrate high accuracy and precision. The shorter inference time for images with plain backgrounds suggests an efficiency gain in simpler visual contexts. These findings underscore the robustness and effectiveness of our integrated model in accurately recognizing student behaviors in varied visual environments.

7 CONCLUSION

In this paper, we propose the Anti-Exam model along with the STEAM-Exam dataset to recognize students’ behavior during AI-assisted scoring examinations in junior high school academic-level

practical examinations. The objective is to uphold exam integrity, and fairness, and support students’ personal development. Our approach integrates MODNet portrait matting techniques with the YOLOv9 model, a highly accurate and fast object detection algorithm. The STEAM-Exam dataset comprises images captured using a real-time AI-assisted scoring device equipped with three cameras, creating a realistic examination environment. Additionally, the dataset includes MODNet-processed images with plain backgrounds. The model’s processing speed varies slightly depending on the background complexity of the images. We evaluated the model and compared the results across YOLOv7, YOLOv8, and YOLOv9 model versions. Our findings indicate that the YOLOv9-E model achieved the highest precision (98.7%), recall (98.7%), mAP50 (99.4%), and mAP50:95 (98.6%). The processing speed for images with backgrounds is 0.3ms for pre-processing, 37.4ms for inference, and 1.5ms for Non-Maximum Suppression (NMS) per image. In contrast, for plain background images, the processing speed is 0.2ms for pre-processing, 35.1ms for inference, and 0.7ms for NMS per image.

We believe that our work will significantly contribute to the artificial intelligence field, particularly in positively impacting the education sector. In future research, we plan to develop an AI-based model for assigning grades in high school academic-level practical

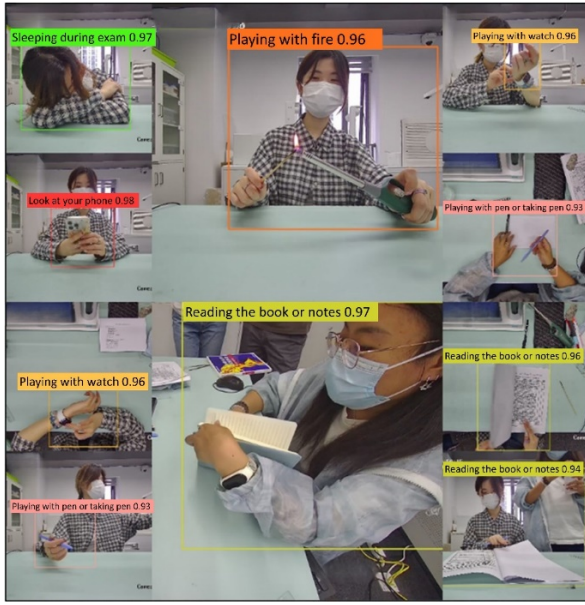


Figure 7: Student's behavior recognition testing results of the proposed model with background

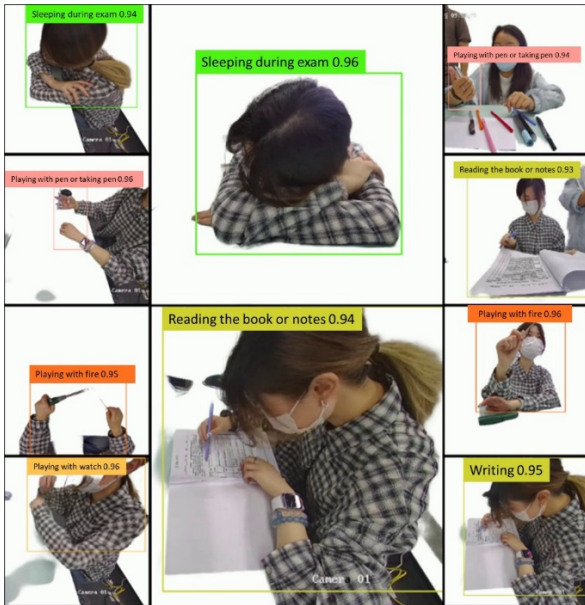


Figure 8: Student's behavior recognition testing results of the proposed model without background

examinations. Additionally, we aim to address the occlusion problem, which leads to inaccurate grading in AI-based grading systems, by utilizing a video motion tracking spatial tracker algorithm. This research offers a robust framework for ensuring exam integrity and fairness while fostering students' personal development. The

integration of advanced AI techniques and real-time data acquisition provides a comprehensive solution for modern educational assessments.

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